Particle Physics 3 Supersymmetry And Grand Unification

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Chapter 1

Supersymmetry & Grand Unification: Lecture 1

These notes follow Leonard Susskind's opening lecture in the supplementary course on supersymmetry and grand unification, curated by LazyingArt LLC. The lecture begins one step earlier than supersymmetry itself. It begins with renormalization: first as a way of throwing away microscopic detail that does not matter for the question at hand, and then as a way of estimating what the discarded short-distance physics must do to the parameters we measure. We will keep that order. The chapter therefore unfolds as the lecture unfolds: coarse-graining, then dimensional analysis, then scalar mass renormalization, then the Higgs hierarchy problem, then the much better behavior of fermions, and finally vacuum energy.

1.1 Renormalization as Coarse-Graining

Susskind opens the course by saying that many of the ideas to come grew out of puzzles in the renormalization of the Standard Model. But he does not begin with counterterms, regularization schemes, or formal perturbation theory. Instead he asks us to remember that physics repeatedly changes its effective language as the scale of interest changes.

Definition 1.1. In the sense emphasized throughout this lecture, renormalization has two ingredients:

1. we eliminate from the description degrees of freedom associated with distances too small, or motions too fast, to matter for the problem we are actually asking;
2. we replace their net effect by effective parameters in a coarser description.

The second half of the subject is dimensional analysis: once the short-distance modes are identified, dimensional analysis often tells us how large their effects must be.

The lecture then climbs, deliberately and patiently, through a ladder of effective descriptions. If we are doing nuclear physics, we may begin with quarks, because quarks are the microscopic constituents. But once we have extracted the masses, spins, and interactions of protons and neutrons, we are free to describe the nucleus as a nonrelativistic many-body system built from those slower composite objects. If we move up to atomic physics, we need even less microscopic information. The detailed

structure of the nucleus mostly drops out; its mass and charge are the main data that survive. Once we know those, we start over again and treat nuclei and electrons as the relevant elementary ingredients.

The same step is taken yet again when we go from atoms to molecules. At each stage we solve just enough of the finer theory to determine the parameters that matter at the next scale, and then we forget how those parameters were produced. The description becomes less microscopic, but more useful. That is the sense in which renormalization is already part of ordinary physics long before we apply the word to quantum field theory.

1.2 Molecules as the Prototype Example

The lecture's first real mathematical example is a pair of atoms, reimagined as two heavy nuclei together with their electrons. The aim is to replace the detailed electronic cloud by an effective force between two slow, heavy objects. Susskind's motivation is physical before it is formal: we would like to describe atoms as particles with forces between them, not as bowling balls permanently accompanied by a swarm of tiny fast flies.

Because relativity is irrelevant at molecular scales, he starts from nonrelativistic quantum mechanics. A cleaned reconstruction of the Hamiltonian written on the board is

$$H = \frac{P_1^2}{2M} + \frac{P_2^2}{2M} + \frac{e^2}{R_{12}} + \sum_i \frac{q_i^2}{2m} + \sum_{i<j} \frac{e^2}{r_{ij}} - \sum_i \left(\frac{e^2}{R_{1i}} + \frac{e^2}{R_{2i}} \right). \tag{1.1}$$

Here P_1, P_2 are the nuclear momenta, M is the nuclear mass, q_i are the electron momenta, m is the electron mass, R_{12} is the distance between the nuclei, r_{ij} is the separation of two electrons, and R_{1i}, R_{2i} are the distances from the i-th electron to the two nuclei.

Remark 1.2. The board image fixes the structure of the Hamiltonian more securely than it fixes every symbol. In particular, the electron momentum notation and the electron-nucleus attraction terms are partly obscured in the frame. The formula above is therefore a cautious transcript-backed reconstruction rather than a literal blackboard transcription.

What matters next is the separation of time scales. The nuclei are heavy. The electrons move much more rapidly. Susskind's picture is that the nuclei move so slowly that, in first approximation, we may nail them down and solve only for the electronic motion in that fixed nuclear background. We group all electron-dependent terms together and treat them as the Hamiltonian of the electrons alone, with R_1 and R_2 held fixed.

We then solve the electronic Schrödinger problem and keep the lowest eigenvalue,

$$E_{\text{el}} = E_{\text{el}}(R_1, R_2). \tag{1.2}$$

This is the ground-state energy of the electrons in the background of stationary nuclei. Since it depends only on the nuclear positions, it can be absorbed into an effective potential for the nuclei:

$$V_{\text{eff}}(R_1, R_2) = \frac{e^2}{R_{12}} + E_{\text{el}}(R_1, R_2). \tag{1.3}$$

The lecture emphasizes the qualitative shape. At very short separation the Coulomb repulsion of the nuclei dominates, so the potential is strongly repulsive. At very large separation the interaction

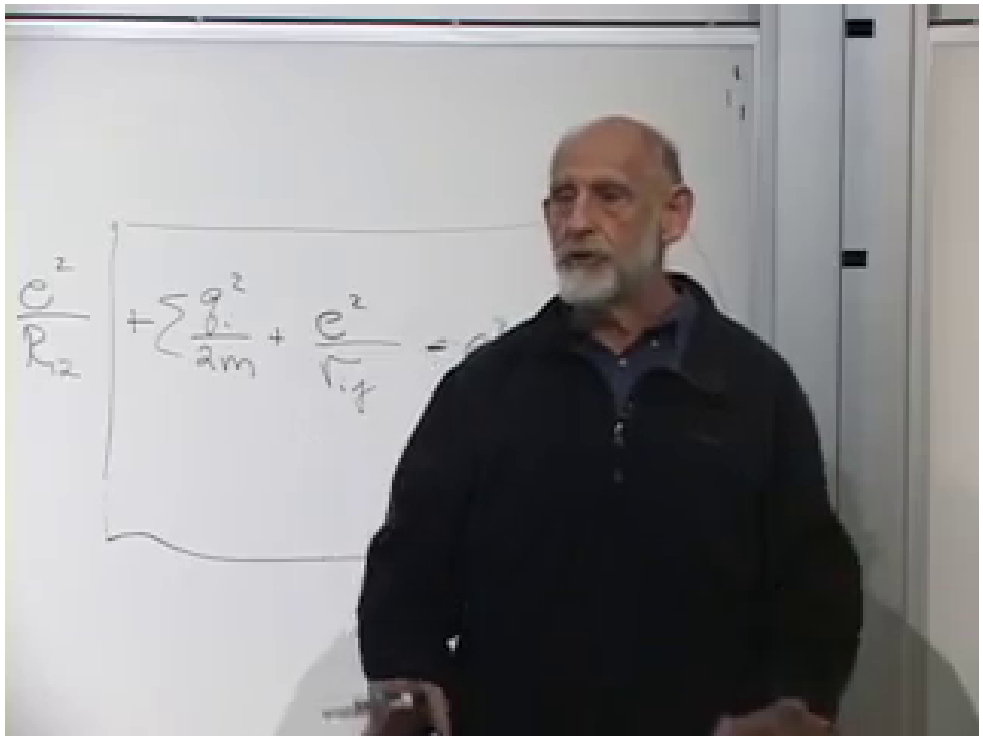

Figure 1.1: Molecular Hamiltonian with the electron sector visually grouped as a subproblem.

becomes weak. Between the two there is an attractive region due to the electrons. Once that effective potential has been found, we no longer track the electronic cloud explicitly. The fast microscopic degrees of freedom have been eliminated and replaced by a renormalized Hamiltonian for the slow ones.

That is the prototype. Susskind explicitly pauses here to say that this, too, is renormalization.

1.3 From Effective Potentials to Field Theory

Only after the molecular example has done its work does the lecture pivot to field theory. The transition is exact: in quantum field theory the modes we wish to eliminate are the very short-wavelength, very high-frequency, very high-energy ones. Small usually goes with fast. The ultraviolet modes are the field-theory analog of the fast electrons in the molecular problem.

Before launching into diagrams, however, Susskind deliberately interrupts the story and inserts a small toolkit. “Before we do that,” he says in effect, “let us do some dimensional analysis.” This is not a digression. It is what will allow the loop estimates to be read off rather than computed in detail.

Setting $c = \hbar = 1$, one independent dimension remains. We may take it to be length. Then

$$[M] = [E] = [p] = L^{-1}, \qquad [x] = [t] = L. \tag{1.4}$$

So mass, energy, and momentum all count as inverse length, while space and time count as length. Once this is fixed, the size of propagators and couplings becomes almost forced upon us.

1.4 Scalar Fields, Propagators, and Mass Renormalization

The lecture now chooses the cleanest possible setting: one scalar field ϕ. On the board the theory is introduced schematically by first writing the local Lagrangian density,

$$\mathcal{L} = (\partial_\mu \phi)^2 - V(\phi), \tag{1.5}$$

and then, on the next line, the action,

$$S = \int d^4x \, \mathcal{L}. \tag{1.6}$$

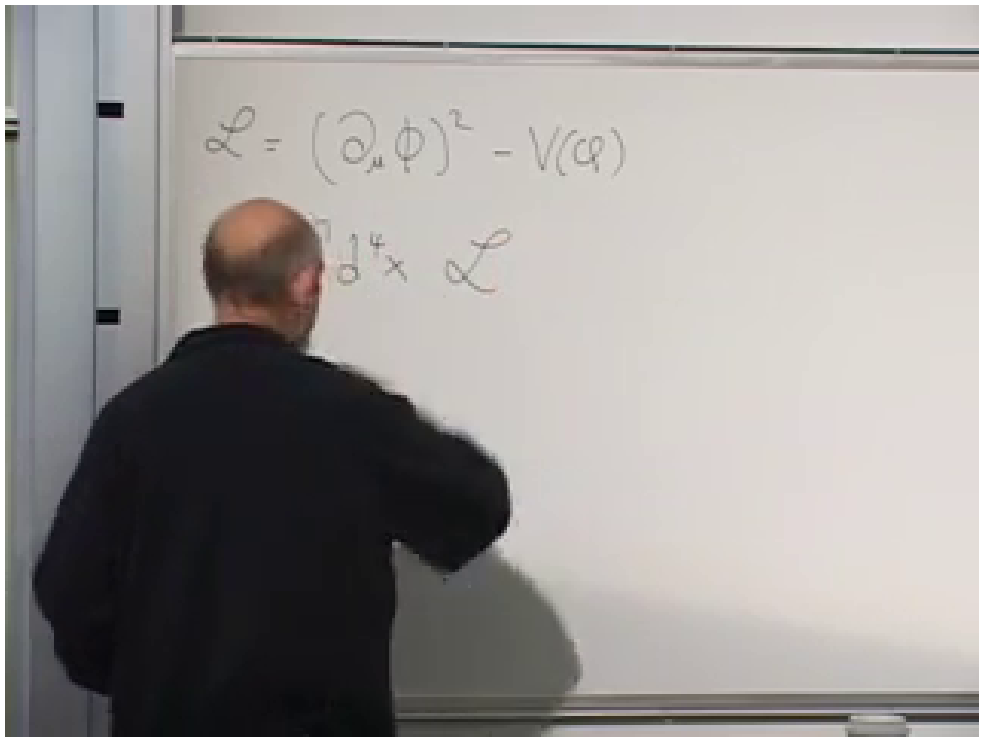

Figure 1.2: The lecture's two-step move from local Lagrangian density to spacetime action.

Remark 1.3. The board writes the scalar theory in a deliberately schematic normalization. We keep that form rather than silently inserting textbook factors of 1/2.

Because $\hbar = 1$, the action is dimensionless. Therefore the Lagrangian density must have dimensions L^{-4}. The kinetic term then fixes the dimension of the field itself:

$$[S] = 1, \qquad S = \int d^4x \, \mathcal{L}, \tag{1.7}$$

$$[\mathcal{L}] = L^{-4}, \qquad [(\partial_\mu \phi)^2] = L^{-4}. \tag{1.8}$$

Since $[\partial_\mu] = L^{-1}$, we obtain

$$[\phi]^2 L^{-2} = L^{-4} \qquad \Longrightarrow \qquad [\phi] = L^{-1}. \tag{1.9}$$

This is exactly the kind of elementary dimensional fact that the lecture wants us to keep in our pocket.

Next the potential is expanded:

$$V(\phi) = \frac{m^2}{2}\phi^2 + g\phi^3 + \lambda\phi^4 + \cdots . \tag{1.10}$$

Since V must also have dimensions L^{-4}, we read off

$$[g] = L^{-1}, \qquad [\lambda] = 1. \tag{1.11}$$

The point is immediate. The coefficients in the potential become the coefficients of Feynman vertices. The cubic term gives a three-point vertex weighted by g. The quartic term gives a four-point vertex weighted by λ. The quadratic term gives a two-point insertion: the lecture draws it as a particle that comes in, is absorbed, and is emitted again from essentially the same point.

The second ingredient is the propagator. Susskind defines it operationally before he defines it algebraically. It is the amplitude to create a particle at one spacetime point and detect it at another. In the lecture this is written as

$$\langle 0|\phi(y)\phi(x)|0\rangle. \tag{1.12}$$

Now the vacuum has no dimensions, but each ϕ has dimensions L^{-1}, so the whole object has dimensions L^{-2}. If the scalar is massless, there is no intrinsic length scale in the problem except the separation of the two points themselves. Therefore the propagator must scale schematically as

$$\langle 0|\phi(y)\phi(x)|0\rangle \propto \frac{1}{(x-y)^2}. \tag{1.13}$$

Here one should read $(x-y)^2$ schematically as the Lorentz-invariant squared separation. The lecture is not deriving a fully normalized Green's function. It is isolating the short-distance behavior. And that behavior is what matters: the propagator blows up as the points approach one another. That is the source of ultraviolet divergence.

Now the lecture asks what the mass term really means in an effective theory. At the microscopic level the bare $m^2\phi^2/2$ term is the simplest local two-point insertion. But if our experimental microscope cannot resolve distances smaller than some cutoff δ, then any short-distance process that imitates a particle coming in and going out again from essentially the same place must be counted as part of the mass term. This is where renormalization enters.

The first example uses the quartic coupling. A particle comes in, goes out, and at the same point a short virtual line is emitted and reabsorbed. If the separation is smeared only over distances of order δ, the propagator contributes $1/\delta^2$, so

$$\delta m^2 \sim \frac{\lambda}{\delta^2}. \tag{1.14}$$

The next example is a higher-order diagram with two quartic vertices. Here we must integrate over the relative separation of the two nearby points. To keep the notation straight, let Δ^μ denote the integration variable, while δ denotes the cutoff distance. Then the lecture's dimensional estimate is

$$\lambda^2 \int d^4\Delta \, \frac{1}{\Delta^6} \sim \frac{\lambda^2}{\delta^2}. \tag{1.15}$$

The logic is simple: three propagators give Δ^{-6}, the four-dimensional measure gives $d^4\Delta$, and the answer can depend only on the lower cutoff δ. So it must again scale like δ^{-2}.

The cubic interaction gives something slightly subtler. Two cubic vertices and two propagators produce

$$g^2 \int d^4\Delta \, \frac{1}{\Delta^4} \sim g^2 \log(\delta \, \mu_{\text{ref}}), \tag{1.16}$$

with a reference scale μ_{ref} inserted only to make the logarithm dimensionless. The lecture treats such logarithms cautiously and loosely; what matters for the present discussion is that logarithms are much weaker than power divergences. When δ is very small, δ^{-2} dominates $\log \delta$.

Putting the structure together, we are led to the schematic relation

$$m_{\text{phys}}^2 \sim m_0^2 + \frac{c_1 \lambda}{\delta^2} + \frac{c_2 \lambda^2}{\delta^2} + c_3 g^2 \log(\delta \, \mu_{\text{ref}}) + \cdots . \tag{1.17}$$

The numerical coefficients and signs are not tracked in the lecture; the dominant message is that the scalar mass receives an infinite series of ultraviolet-sensitive corrections. And the same is true of the other couplings. The coefficient λ itself is renormalized by four-point diagrams, typically logarithmically, although Susskind stops short of turning that into a full study of beta functions.

1.4.1 Question & Answer

Question. What are we actually measuring: the input parameter or the renormalized one?

Answer. We measure the renormalized one. The point of the cutoff description is precisely that we refuse to resolve all of the very short-distance structure. Once we do that, every unresolved process that imitates a local two-point interaction is lumped into the coefficient we call the mass term. The laboratory therefore does not see the pristine m_0^2. It sees the effective coefficient after all the short-distance fluctuations we have chosen not to resolve have been summed into it.

1.5 From Scalar Renormalization to the Higgs Hierarchy Problem

At this point the lecture turns the scalar example into a crisis. For a scalar particle the additive corrections are present even if the bare mass vanishes. That is what makes the Higgs field dangerous. The lecture therefore asks us to recall the Higgs potential in its broken-symmetry form:

$$V(\phi) = -\frac{\mu^2}{2}\phi^2 + \lambda\phi^4. \tag{1.18}$$

Because the board normalization is schematic, the minimum is quoted in the lecture in the convention used there as

$$\phi_{\text{min}}^2 = f^2 = \frac{\mu^2}{\lambda}. \tag{1.19}$$

The key point is the scale, not the harmless convention-dependent factors. The vacuum expectation value is already known from electroweak physics:

$$f \sim 200\,\text{GeV}. \tag{1.20}$$

So unless λ is made implausibly tiny or huge, μ is also of electroweak order.

Now compare that with the ultraviolet-sensitive scalar corrections. If we trust quantum field theory all the way down to a scale as short as the Planck length, then in GeV units the lecture estimates

$$\frac{1}{\delta^2} \sim 10^{38}\,\mathrm{GeV}^2. \tag{1.21}$$

This is the point where the tone of the lecture changes. The scalar story is no longer just a technical example of renormalization. It becomes bad behavior. We want a Higgs parameter of electroweak size, but the short-distance contributions are incomparably larger. The precise number of decimal places shifts a bit in the spoken lecture, and the signs of the various contributions are left untracked, but the central point is stable: a huge collection of ultraviolet contributions must cancel to extraordinary precision in order to leave behind a weak-scale mass parameter.

That is the hierarchy problem, or fine-tuning problem, of the Higgs sector. It is not just that the weak scale is small compared with the Planck scale. It is that, in a scalar theory, the small physical parameter appears as the residue of a delicate cancellation among individually huge pieces.

1.5.1 Question & Answer

Question. Must the bare mass term cancel all the short-distance corrections?

Answer. If we insist on keeping a very short-distance cutoff while also insisting that the physical Higgs parameter remain at the electroweak scale, then yes: the bare term must conspire with the radiative corrections so that the large ultraviolet-sensitive pieces almost exactly cancel. That is what the lecture means by fine-tuning. The problem is not that a correction exists. The problem is that a very small measured answer seems to require a cancellation among many much larger terms.

1.6 Why Fermions Behave Better

The lecture then deliberately changes tempo and asks the obvious question: if scalar masses are so unstable, why is the electron not equally disastrous? The answer comes from the structure of the fermion mass term,

$$m\bar{\psi}\psi. \tag{1.22}$$

This term is linear in the fermion mass, not quadratic. More importantly, it flips chirality. It takes a left-chiral fermion into a right-chiral one, or vice versa. Susskind sometimes phrases this in helicity language in the lecture, but the mathematically safer language here is chirality.

The lecture now contrasts three kinds of vertices. Gauge-boson emission preserves chirality: $L \to L$ and $R \to R$. Scalar emission flips chirality: $L \leftrightarrow R$. The mass insertion $m\bar{\psi}\psi$ also flips chirality. The right-hand chalk sketches are mnemonic rather than literal Feynman graphs, but the underlying logic can be summarized schematically as follows.

Now suppose the electron begins massless. Can photon emission and reabsorption generate a mass? No. Every gauge interaction preserves chirality, so no chain of such processes can turn left into right. Can scalar emission and reabsorption do it? Again no. A scalar emission flips chirality, but the corresponding reabsorption flips it back. Since the emitted scalar must be reabsorbed, the total number of scalar vertices is even, and the net effect cannot be a left-right transition. So neither gauge-boson loops nor scalar loops generate an additive fermion mass from zero.

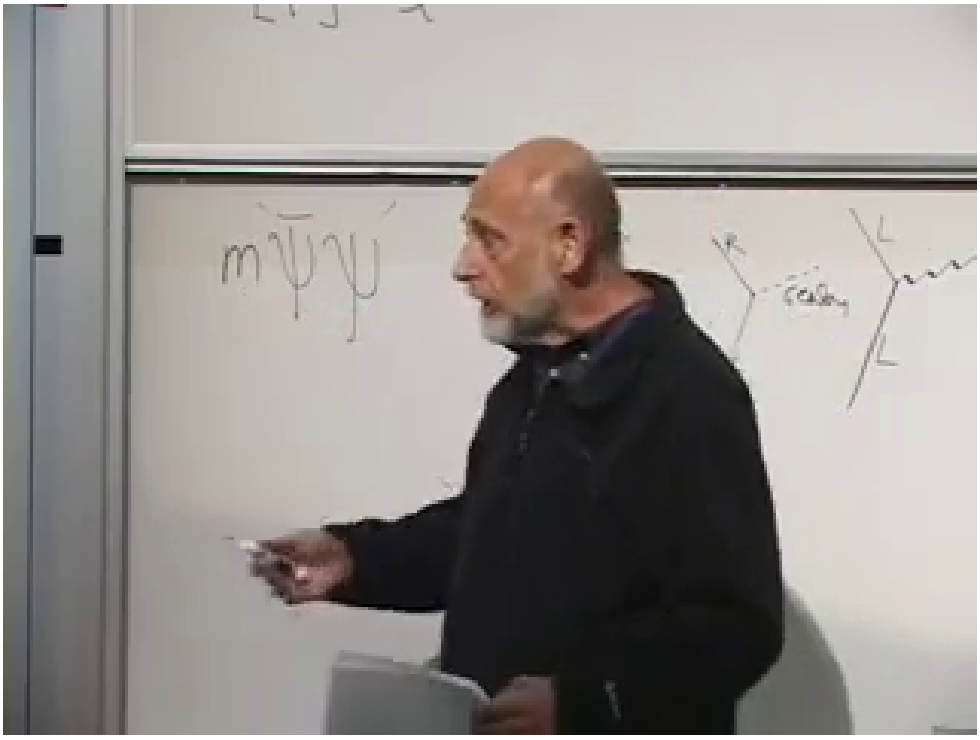

Figure 1.3: The fermion mass term on the board, together with the lecture's chirality mnemonics.

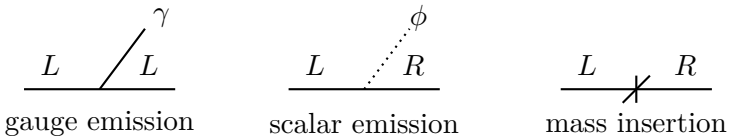

Figure 1.4: Schematic chirality mnemonics: gauge emission preserves chirality, while scalar emission and mass insertion flip it.

To get a correction to the fermion mass, one must already insert a mass somewhere in the diagram. Then the correction is proportional to the starting mass itself. That is the lecture's crucial point. Schematically,

$$m_e = m_0 + O(e^2 m_0) + O(e^4 m_0) + \cdots . \tag{1.23}$$

Therefore if $m_0 = 0$, the electron stays massless. If m_0 is small, all the corrections remain small. One may still ask why the electron mass is numerically small compared with some deeper scale, but one is not forced into the same kind of catastrophic cancellation that afflicted the scalar.

Remark 1.4. The lecture slides back and forth between chirality and helicity in a physically intuitive way. For the purpose of writing the argument cleanly, chirality is the better variable. In the relativistic limit the lecture's helicity-based intuition remains a useful picture.

Gauge bosons behave in the same protected way for the purpose at hand: if they begin massless, they remain massless to all orders. That is why the lecture insists that the Standard Model does not contain a separate fine-tuning problem for every light particle. The truly severe fine-tuning is the one tied to the Higgs field. If that one is solved, the rest of the light spectrum falls into place.

1.6.1 Question & Answer

Question. Why doesn't the electron have the same fine-tuning problem?

Answer. Because the electron mass is protected against additive renormalization. Starting from zero, neither gauge interactions nor scalar emission-and-reabsorption can generate the chirality-flipping two-point structure that defines a fermion mass. Any nonzero correction already contains one explicit mass insertion, and is therefore proportional to the original mass itself. A small fermion mass may remain a fact in need of explanation, but it is not produced by canceling huge ultraviolet contributions against each other.

1.7 Vacuum Energy and the Supersymmetry Outlook

The lecture closes by widening the frame once more. There is another quantity that renormalization shifts: the energy of the vacuum itself. At first one is tempted to say, with Susskind, "who cares?" In ordinary nongravitational physics only energy differences matter. If the vacuum energy is shifted by a constant amount, that seems irrelevant so long as adding a particle adds the right extra amount of energy.

But gravity does care. Empty space gravitates if it carries energy. So the renormalization of the vacuum energy becomes physically significant once the gravitational field is allowed to respond to it.

In ϕ^4 theory the simplest vacuum-to-vacuum diagram contains no external particles. It consists only of a quartic vertex together with short propagators that begin and end within the unresolved region. The lecture's estimate is then

$$\delta\rho_{\text{vac}} \sim \frac{\lambda}{\delta^4}. \tag{1.24}$$

More complicated vacuum diagrams typically preserve the same δ^{-4} behavior in this schematic discussion. So vacuum energy is also highly sensitive to short-distance physics.

The lecture therefore ends with two fine-tuning problems, not one. The Higgs mass is one. The vacuum energy is the other, and it is even more severe once gravity is included. Supersymmetry is introduced as a possible response to the first of these. It is not presented here as a cure for the cosmological-constant problem.

1.7.1 Question & Answer

Question. Does the fine-tuning depend on the arbitrary cutoff δ?

Answer. The contribution does depend on how far down in distance we trust the theory, but that is not an empty arbitrariness. We choose δ to be the shortest scale at which we still believe the present quantum field theory makes sense. Then the accumulated contribution from all scales above that point is already there and must be accounted for. In that sense the estimate is a lower bound on the size of the problem. If the theory remains valid to even shorter distances, the scalar fine-tuning can only become worse. The lecture mentions scales as small as unification, around 10^{16} GeV, as a conservative benchmark, with the Planck scale making the tension sharper still.

Two late clarifications from the lecture belong here. First, a scalar particle is a spin-zero particle; the Higgs is the central example in this discussion. Gauge bosons, by contrast, are spin-one particles. Second, although the simplest fermion and boson vacuum diagrams come with opposite signs, cancellation is not automatic. To cancel the vacuum energy naturally, masses and couplings would have to be matched with extraordinary precision. That is the kind of matching supersymmetry tries to impose in the Higgs sector, which is why it arrives here as the next subject of the course.

1.8 Summary

The lecture begins by teaching us how to think about renormalization before teaching us how to compute with it. We first integrate out fast electrons in a molecular problem and replace them by an effective potential for slow nuclei. We then carry the same idea into quantum field theory, where short-distance modes renormalize the coefficients in the Lagrangian. Dimensional analysis shows why scalar propagators are singular at short distance and why scalar mass terms are especially vulnerable to large additive corrections. That lesson becomes the Higgs hierarchy problem once the cutoff is taken seriously. Fermions and gauge bosons behave better because their masses are protected when they start from zero. Finally the lecture widens from the Higgs problem to the renormalization of vacuum energy, and ends by presenting supersymmetry not as a finished solution, but as the natural next attempt to tame the Higgs-sector fine-tuning.

Chapter 2

Supersymmetry and Grand Unification: Lecture 2

This lecture does not begin by defining supersymmetry. It does something more characteristic. It backs up. Before bosons and fermions can be related to one another, we are asked to remember what is already peculiar about fermions. Before that, we are asked to remember something peculiar about rotations themselves. The chapter should therefore be read in the same order in which the lecture unfolds: rotations, signs, exchange, loops, cancellations, broken supersymmetry, and only then the question of what an exactly supersymmetric world would look like.

The payoff is twofold. On the mathematical side, opposite signs for bosonic and fermionic loops create the possibility of cancellation. On the physical side, exact supersymmetry would radically alter ordinary matter. Between those two observations lies the lecture's real theme: supersymmetry is interesting precisely because exact supersymmetry is not the world we live in.

2.1 Rotations, 2π, and the Topological Prehistory of Supersymmetry

"Actually before we talk about supersymmetry," Susskind says, "we should talk again about fermions and bosons." Then he backs up once more: before fermions, rotations in space. The familiar statement that a rotation by 2π is the same as no rotation at all is, in the topological sense relevant here, too naive. He demonstrates the point physically with his arm, and then translates it into the box-and-ball picture.

We imagine a ball in a box, connected to the walls by strings. The walls represent spatial infinity, so the box is held fixed. A rotation of the ball by 2π tangles the strings, and the tangle cannot be removed if the ball is not rotated further. A rotation by 4π, however, can be undone by moving the strings around the ball. He jokingly says he has already proved the theorem "by hand-waving." That is enough for the lecture. The point is not a formal homotopy proof; the point is the qualitative fact that 2π is not topologically trivial, while 4π is.

This is the geometric memory that later reappears in fermions. In one of the discussion remarks Susskind even says, roughly, that bosons behave as though there were no strings attached to the wall, whereas fermions remember that a 2π rotation has taken place.

Now we freeze the position of a particle and look only at its spin state. Following the lecture, we may temporarily call that state $|S\rangle$; more generally we write $|\psi\rangle$. The simplest possibility is that a 2π rotation acts by multiplication with a number:

$$R(2\pi)\,|\psi\rangle = \xi\,|\psi\rangle\,. \tag{2.1}$$

If we perform the same rotation twice, then

$$R(4\pi)\,|\psi\rangle = \xi^2\,|\psi\rangle\,. \tag{2.2}$$

But the topological statement was precisely that 4π is equivalent to doing nothing. Therefore

$$R(4\pi)\,|\psi\rangle = |\psi\rangle\,, \qquad \xi^2 = 1, \qquad \xi = \pm 1. \tag{2.3}$$

So the state may come back to itself after a 2π rotation either with a plus sign or with a minus sign:

$$R(2\pi)\,|\psi_{\text{boson}}\rangle = +\,|\psi_{\text{boson}}\rangle\,, \qquad R(2\pi)\,|\psi_{\text{fermion}}\rangle = -\,|\psi_{\text{fermion}}\rangle\,. \tag{2.4}$$

That already separates bosons from fermions in a very stark way. The lecture emphasizes that such a rotation is not merely formal. One can implement it in the laboratory by letting a spin precess in a magnetic field, or by rotating the magnetic field itself. The mathematical distinction is therefore physically meaningful, if only we can find the right way to see it.

2.2 Spin States Under Rotation and the Interference Test of a Sign Change

At first sight a minus sign seems like no physics at all. If we rotate every copy of a wavefunction by 2π and the result is just $-\Psi$, then every observable constructed from $\Psi^*\Psi$ is unchanged. That is exactly the paradox the lecture wants to raise before it resolves it.

2.2.1 Question & Answer

Can the minus sign from a 2π rotation ever be observed?

Yes, but only as a relative sign.

The lecture answers with a two-path interference experiment. A single electron is prepared in a superposition of two branches, one going through an upper path and one through a lower path. If the two branches are ψ_1 and ψ_2, then before anything is rotated the total wavefunction is

$$\Psi = \psi_1 + \psi_2. \tag{2.5}$$

Now imagine that while the electron is trapped in one of the two regions, we rotate only one branch by 2π. For a fermion, that branch picks up a minus sign. If it is the second branch that is rotated, the recombined state becomes

$$\Psi' = \psi_1 - \psi_2. \tag{2.6}$$

So the physical question is not whether Ψ differs from $-\Psi$. The physical question is whether

$$\psi_1 + \psi_2 \qquad \text{or} \qquad \psi_1 - \psi_2 \tag{2.7}$$

emerges after recombination.

If the two returning wave packets are symmetric by reflection about the center of the screen, then the odd combination must vanish there:

$$\Psi_{\text{odd}}(0) = 0. \tag{2.8}$$

That is the cleanest way to state the lecture's argument. A relative minus sign shifts the interference pattern; maxima become minima, and the electron is never found at the symmetry point. On the other hand, if both branches are rotated together, the full state changes only by an overall sign, and then nothing observable happens:

$$|\Psi|^2 = (-\Psi)^*(-\Psi). \tag{2.9}$$

This is why the sign under 2π is real physics but only in a relational way. Rotating half the experiment relative to the other half is like rotating the ball relative to the box: memory can then be detected. Rotating the whole experiment together is like rotating ball and box together: the effect disappears. And if the relative rotation were 4π, the extra sign would disappear as well.

The lecture also makes a useful contrast here. The special role of 2π is that it changes only the sign of a fermion state while leaving the spin orientation effectively the same. A generic rotation changes the spin state more substantially. Then the two branches are no longer the same object up to a sign, and the interference pattern is partially washed out or even destroyed. A 180° flip, for example, destroys it completely.

2.3 Bosons, Fermions, and Exchange Symmetry

Having extracted one distinction between bosons and fermions from rotations, the lecture turns to a second one: exchange symmetry. This shift is not accidental. The next target is a sign rule for Feynman loops, and that rule depends on what happens when identical particles are interchanged.

Let $\Psi(1,2)$ be the two-particle wavefunction, and let P_{12} denote the operator that exchanges particles 1 and 2. Then the standard statements are

$$P_{12}\Psi(1,2) = +\Psi(1,2), \qquad \Psi(1,2) = +\Psi(2,1) \qquad \text{for bosons,} \tag{2.10}$$

$$P_{12}\Psi(1,2) = -\Psi(1,2), \qquad \Psi(1,2) = -\Psi(2,1) \qquad \text{for fermions.} \tag{2.11}$$

The lecture phrases this more directly: bosonic wavefunctions are symmetric, fermionic wavefunctions are anti-symmetric.

The board evidence is especially useful here because Susskind is speaking in ket language at this moment, not only in coordinate language.

The only symbol that is really clear on the board is $|S\rangle$. So the cleaned equations have to be treated as careful reconstructions rather than verbatim board transcriptions. The mathematically appropriate statements are

$$\Psi(1,2) = -\Psi(2,1), \tag{2.12}$$

$$P_{12}\,|\psi\rangle = -\,|\psi\rangle\,. \tag{2.13}$$

The second line is the lecture's ket-language version of the same fact. It is this minus sign under exchange, not some abstract slogan about statistics, that is about to reappear in the sign of a closed fermion loop.

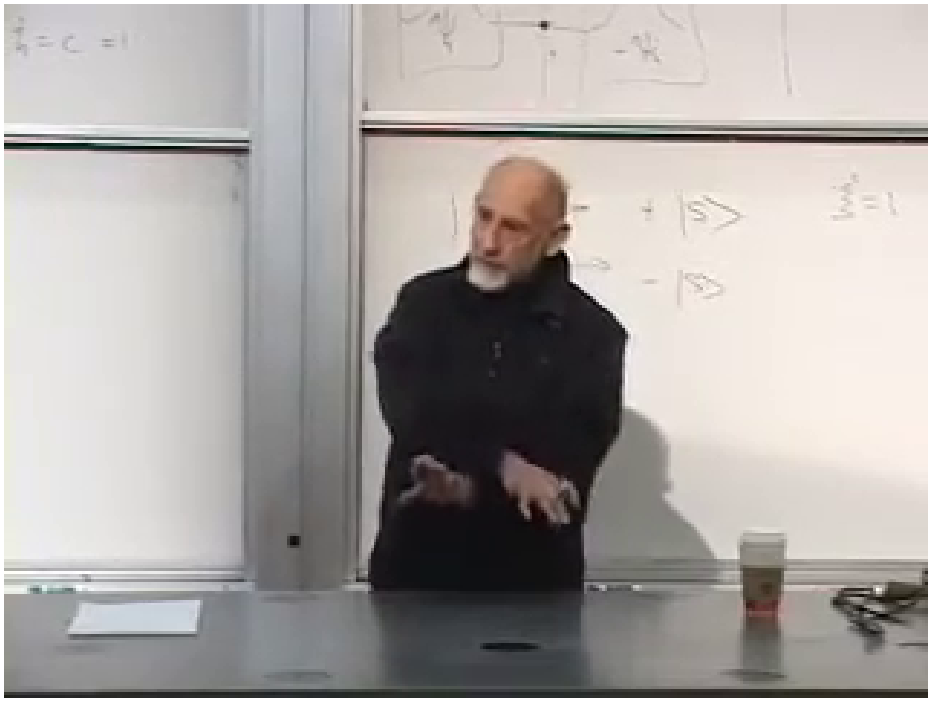

Figure 2.1: Fermion anti-symmetry and state-label sketch. The screenshot is kept because it preserves the board layout and the ket-style presentation, even though only $|S\rangle$ is securely legible.

2.4 Why a Closed Fermion Loop Carries a Minus Sign

We now come to what Susskind calls "yet another question": a question about very simple Feynman diagrams. These are vacuum diagrams with no external lines, just a closed loop in empty space. One may think of the loop either as a particle moving around a closed curve in spacetime or, more physically, as pair creation followed by annihilation. The precise integral associated with the diagram is not the point here. The point is the sign.

2.4.1 Question & Answer

Why does one closed fermion loop come with a minus sign?

The lecture gives an elegant proof that uses nothing heavier than exchange symmetry.

Begin with two disconnected loops. Since disconnected diagrams multiply, the sign of the two-loop diagram is the square of the sign of the one-loop diagram:

$$\text{sign}(\text{two disconnected loops}) = \big(\text{sign}(\text{one loop})\big)^2 = +1. \tag{2.14}$$

So two disconnected loops always carry an overall plus sign, regardless of whether one loop by itself is positive or negative.

Now cut both loops at a common time slice. At that instant we see two identical particles. There are then two possibilities: reconnect them without exchanging the particles, or interchange the two particles before reconnecting them. The first operation gives back the original two disconnected loops. The second changes the amplitude by the exchange sign:

1. For bosons, exchanging the two particles contributes no extra sign.
2. For fermions, exchanging the two particles contributes a minus sign.

But after this interchange and reconnection, what had been two loops has become one single loop. Therefore the sign of the one-loop diagram must be

$$\text{sign(one closed boson loop)} = +1, \qquad \text{sign(one closed fermion loop)} = -1. \tag{2.15}$$

That is the whole proof. It is not a path-integral derivation, and the lecture does not want one here. It wants the elementary mechanism to be visible: the extra minus sign comes from exchanging identical fermions. The consequences are immediate. Bosonic and fermionic loops contribute with opposite signs to quantum corrections. We have finally reached the first real mathematical doorway into supersymmetry.

2.5 Vacuum Energy, Boson Self-Energy, and the Zeroth-Order Supersymmetry Idea

The first place Susskind uses the loop-sign result is the vacuum energy. A closed boson loop contributes with one sign; a closed fermion loop contributes with the opposite sign. Since the observed vacuum energy is astonishingly small compared with the naive quantum field theoretic estimate, the lecture pauses over the fact that cancellation is at least possible.

All things being equal, meaning the same mass and the same relevant quantum numbers, a boson and a fermion would give equal contributions in magnitude and opposite contributions in sign. Schematically,

$$m_{\text{partner}} = m_{\text{particle}}, \qquad Q_{\text{partner}} = Q_{\text{particle}}, \tag{2.16}$$

and then one may hope for

$$\Delta E_{\text{vac}}^{(b)} + \Delta E_{\text{vac}}^{(f)} = 0. \tag{2.17}$$

The lecture is careful not to oversell this. It does not say that the cosmological constant problem has been solved. It says only that the sign structure creates room for cancellation, which is already something important.

Then comes one of Susskind's characteristic pivots: "another thought to keep in mind." Vacuum energy is not the main issue. The sharper question is whether the same logic can help with the self-energy of bosons, especially Higgs-like scalars. Bosonic loop corrections by themselves drive scalar masses upward. Fermions, by contrast, can be protected against such enormous corrections. So if a theory forced bosons and fermions to travel in closely matched pairs, one could imagine transferring that protection to the bosons. In schematic form,

$$\Delta m_B^2\big|_{\text{boson loop}} + \Delta m_B^2\big|_{\text{fermion loop}} \approx 0. \tag{2.18}$$

Exact partner matching would give exact cancellation. Near matching would give only approximate cancellation, but approximate cancellation is enough if the goal is merely to keep the Higgs sector from running away to enormous scales.

After the break, the lecture rephrases the whole idea "to zeroth order," not in the strict mathematical sense of an expansion, but as a rough first picture. The rough picture is that every particle in nature comes with a superpartner of the opposite statistics. Here "statistics" is exactly Susskind's word for

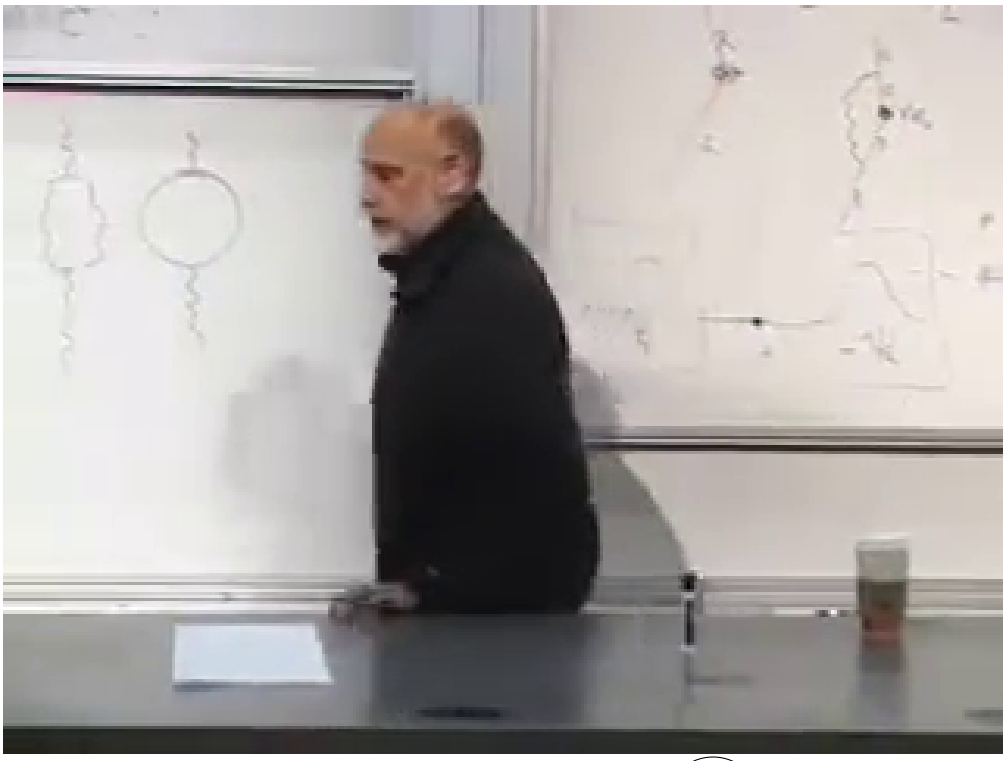

Figure 2.2: Loop-diagram comparison in the supersymmetry discussion. The redraw is intentionally schematic: the left-hand side shows simple loop topologies, while the right-hand side suggests loop corrections to propagating particles.

fermion-ness versus boson-ness. Supersymmetry, at this stage of the lecture, is the hypothesis that the spectrum is doubled in just this way.

The board at this point is valuable mainly as layout evidence: simple loops on the left, more elaborate loop-corrected structures on the right.

The lecture still has not defined the supersymmetry algebra. It is doing something more primitive and, pedagogically, more effective. It is giving us a reason to want such a symmetry in the first place.

2.6 Broken Supersymmetry, Naturalness, and Experimental Reach

At this point the lecture turns from motivation to reality. Exact mass matching would give exact cancellation, but exact mass matching is not how nature looks. We know this at once from experiment. If there were a bosonic partner of the electron with the same charge and the same mass, it would already have shown up in ordinary production processes. An electron-positron pair can annihilate into a virtual photon, and if the partner were truly degenerate with the electron, the

virtual photon could produce the partner pair just as easily:

$$e^-e^+ \to \gamma^* \to \tilde{e}^-\tilde{e}^+. \tag{2.19}$$

We do not see that. So exact supersymmetry is not the world we live in.

2.6.1 Question & Answer

If supersymmetry helps, why have we not seen the partners?

Because the symmetry, if it exists in nature, must be broken.

The lecture insists on two points at once. First, the masses of particles and superpartners are certainly not equal. Second, the splittings cannot be arbitrarily large if supersymmetry is supposed to do the job for which it was invented. Exact degeneracy would produce exact cancellation. Near degeneracy gives approximate cancellation. But if the partner masses were pushed all the way up to some enormous scale, then the cancellations would be poor and the Higgs self-energy problem would simply return in a slightly disguised form.

This is the naturalness argument in the form the lecture wants it. If superpartners were as heavy as the unification scale or the Planck scale, their loops would drag the Higgs sector upward almost as badly as if there were no supersymmetry at all. So a supersymmetry that protects the Higgs mass requires that the particle-superpartner splittings not be too large. Susskind gives a lecture-era estimate of order a few hundred GeV as the scale beyond which the cancellation becomes too poor to be convincing. The exact number should be taken historically rather than timelessly; the logical point is the important one.

The lecture also rejects a merely aggregate cancellation, where the sum of all fermionic effects accidentally cancels the sum of all bosonic effects. That would be just another fine-tuning. The supersymmetric hope is a structured cancellation, enforced by a symmetry, not a miraculous bookkeeping accident.

Here the line-style contrast between ordinary loops and partner loops becomes visually explicit.

The lecture then sharpens the matching requirement. It is not enough that masses be comparable. Couplings must match as well. Once external photons are attached to the loop, cancellation requires that the partner carry the same electric charge as the original particle:

$$Q_{\text{partner}} = Q_{\text{particle}}. \tag{2.20}$$

Otherwise the two diagrams do not have the same numerical coefficient and the cancellation fails.

Experimentally, there are then two fronts. One may look directly for superpartner production in sufficiently energetic collisions. Or one may look indirectly for their virtual effects in high-precision electroweak processes. The lecture emphasizes that electron-positron machines are cleaner for the second purpose than hadron colliders, because protons are composite and the collision environment is correspondingly messy. The high-precision data do constrain supersymmetry, but in the lecture's discussion they do not yet rule it out.

Susskind also remarks that most superpartners would not be expected to sit around forever. They should decay, perhaps cascading down to a lightest superpartner. In many models that lightest state is neutral and may be stable. But the main phenomenological message of the lecture is simpler: if supersymmetry is to protect the Higgs sector, the new particles must lie not too far beyond experimental reach.

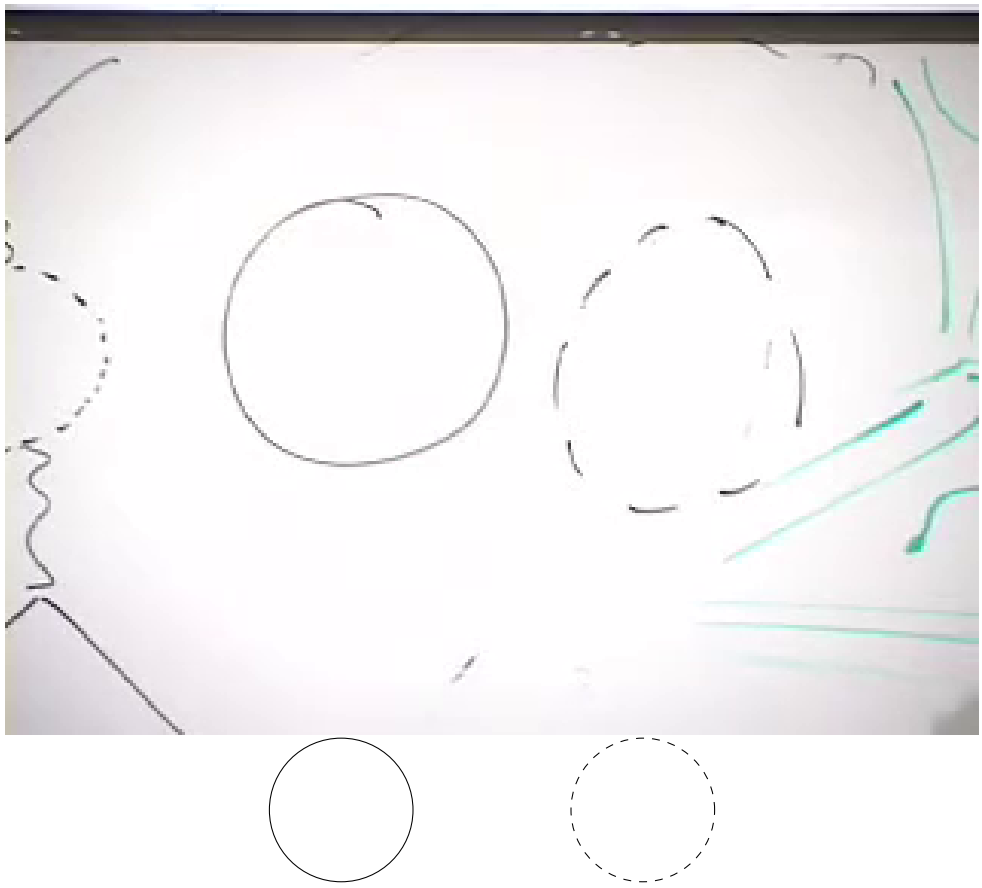

Figure 2.3: Solid and dashed loop topologies. The essential content is the side-by-side comparison of ordinary and superpartner loop contributions with different statistics and therefore opposite signs.

2.7 Exact Supersymmetry Would Destroy Ordinary Chemistry

Before closing, the lecture fixes the relation between the spins of particles and their superpartners. Bosons are paired with fermions, fermions with bosons, and the spin changes by half a unit:

$$j_{\text{superpartner}} = j \pm \frac{1}{2}. \tag{2.21}$$

Thus a gauge boson has a spin-$\frac{1}{2}$ partner, while an ordinary spin-$\frac{1}{2}$ fermion such as the electron has a scalar partner. This is also where the familiar names appear. Bosonic particles acquire fermionic partners with "-ino" names such as photino and Higgsino, whereas fermions acquire bosonic partners such as the selectron and the squark. The lecture also notes that extra scalar fields raise the danger of additional spontaneous symmetry breaking, which would generally be unwelcome and must be controlled.

There is a brief aside about lower dimensions: in two spatial dimensions the boson-fermion dichotomy is generalized, and one finds anyons with more general spin-statistics behavior. The lecture does not pursue that topic further. Instead it asks the question with which it really wants to end.

2.7.1 Question & Answer

What would the world be like if supersymmetry were exact?

Very different. In particular, ordinary chemistry would fail.

Here "exact" means exact degeneracy between particles and superpartners. If that were true, the electron would have a bosonic partner with exactly the same mass and charge, and the photon would have a massless fermionic partner, the photino:

$$m_\gamma = 0 \qquad \Rightarrow \qquad m_{\tilde{\gamma}} = 0. \tag{2.22}$$

Now consider an atom with more than two electrons. In hydrogen and helium the available electrons occupy the lowest orbit. In lithium and heavier atoms, outer electrons sit farther out because the low-lying states are blocked by the Pauli exclusion principle. The outer electron wants to fall inward, but ordinary Fermi statistics forbids it.

In an exactly supersymmetric world there is a new escape route. The outer electron can emit a photino and turn into a selectron:

$$e^-_{\text{outer}} \to \tilde{e}^- + \tilde{\gamma}. \tag{2.23}$$

That is the key move in the lecture's final argument. The emitted particle is a fermion, so what remains must be a boson. Once the electron has turned into a selectron, Pauli blocking no longer applies. The selectron can drop to the lowest orbit.

The force of the argument is cumulative. It is not only one outer electron in a lithium-like atom that would do this. Valence electrons more generally would have the same route available to them. They would convert into bosonic selectrons and fall inward. Instead of ordinary shell structure, one would have something closer to a Bose condensate of selectrons gathered near the nucleus. Whatever such a world might be, it would not support familiar atoms, familiar valence structure, or familiar chemistry.

The lecture even anticipates the obvious objection: perhaps the coupling for this transition is tiny. But here there is no escape of that sort. The relevant coupling is electromagnetic. In the

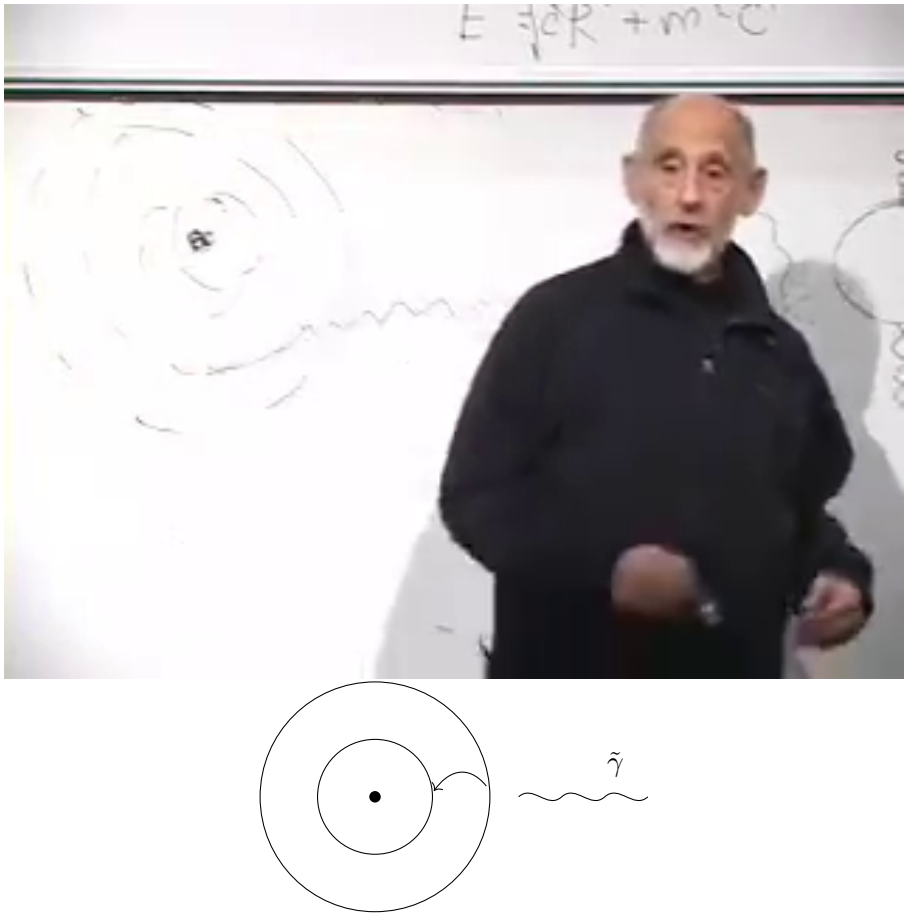

Figure 2.4: Radiative drop to a lower orbit. The board shows emitted radiation and an inward transition; the photino label in the redraw is supplied from the transcript rather than from a legible board inscription.

lecture's rough estimate, a lithium-like atom would not linger in its ordinary form for long at all; the transition would be rapid on atomic timescales.

So the closing lesson is not merely that exact supersymmetry is experimentally excluded. It is that exact supersymmetry would make a world unlike ours. Susskind's answer to the question "why is the symmetry broken?" is correspondingly blunt: if it were not broken, we would not be here to ask. That is not a complete theory of supersymmetry breaking, but it is the right final physical punch.

2.8 Summary

The lecture reaches supersymmetry by refusing to define it too early. First we learn that a 2π rotation is not topologically as innocent as it seems, whereas a 4π rotation is. Then we learn that a spin state may therefore come back under 2π with a sign, and that fermions pick up -1 while bosons do not. Then the sign is made observable through interference, where relative minus signs shift the pattern even though global ones do not.

Only after that does the lecture turn to exchange symmetry and then to Feynman loops. Antisymmetry under exchange implies that a single closed fermion loop comes with a minus sign, whereas a boson loop comes with a plus sign. From there the supersymmetric idea becomes natural: if bosons and fermions occur in sufficiently matched pairs, their quantum corrections can cancel. That possibility first appears in the vacuum energy and then, more importantly for the lecture, in the self-energy of bosons.

Nature, however, does not realize the exact version of the idea. The partners are not degenerate with the known particles, so supersymmetry, if it exists, must be broken. But it cannot be broken too badly if it is to protect the Higgs sector from enormous radiative corrections. And if it were not broken at all, ordinary chemistry would collapse. That is the balance the lecture leaves us with: supersymmetry is interesting because it can cancel dangerous quantum effects, and it must be broken because an exactly supersymmetric world would not look anything like ours.

Chapter 3

Divergences, Propagators, and the Supersymmetric Motive

This lecture was supposed to begin supersymmetry itself. Instead, it does something more deliberate: it backs up and rebuilds the need for supersymmetry from the problem of ultraviolet divergences, especially the renormalization of masses. That is the right order. Before we talk about a powerful new symmetry, we should understand what sort of damage needs repairing. The chapter therefore unfolds in the same order as the lecture: first effective Lagrangians and propagators, then the short-distance behavior of scalar and fermion fields, then the simplest divergent loop corrections, and only at the end the first real glimpse of what supersymmetry is supposed to accomplish.

3.1 Why We Are Still Talking About Divergences

A great deal of particle physics is the computation of Feynman diagrams. To say that, however, is not yet to say what Feynman diagrams are for. The lecture begins by reminding us that they serve at least two rather different purposes.

First, they compute scattering amplitudes. Particles come in, particles go out, and we sum the diagrams that contribute to the probability amplitude. Second, and for the present lecture more importantly, they help us construct an effective Lagrangian. In that setting the point is not to write the exact microscopic theory in a different notation. The point is to write an approximate theory that is effective because it has been stripped of short-distance complications that we do not yet know how to calculate.

That is why a cutoff enters almost immediately. An effective Lagrangian is typically accompanied by either

- a smallest distance, or
- a largest momentum.

The lecture keeps both descriptions alive from the start, because the same ultraviolet problem will soon be read in both languages.

3.2 Propagators in Position Space and Momentum Space

The propagator is the bridge between those two descriptions. If a particle is inserted at one spacetime point and later detected at another, the corresponding amplitude depends on the separation between the two points. To keep the notation unambiguous on the page, we shall write

$$\Delta^\mu = y^\mu - x^\mu \tag{3.1}$$

for the spacetime separation, reserving δ for the ultraviolet cutoff length.

For a scalar field, the propagator is used in the lecture schematically as both an amplitude from x to y and as a vacuum expectation value:

$$G_\phi(x, y) \sim \langle 0|\phi(x)\phi(y)|0\rangle. \tag{3.2}$$

The exact time-ordering conventions are not the point here. What matters is that the propagator depends on x and y only through their difference.

Now the lecture makes a move that is conceptually simple but structurally crucial. A closed loop may be regarded in either of two ways:

- as a propagator from a point back to itself;
- as an integral over all momentum that can circulate around the loop.

That is the same diagram seen from position space and momentum space. Once one sees this, the later discussion of divergences becomes almost inevitable.

3.2.1 Question & Answer

Question. Why are short-distance divergences the same thing as high-momentum divergences?

Answer. Because momentum is inverse wavelength. Very large momentum means very short wavelength, and very short wavelength means very fine spatial resolution. Thus a singularity when $\Delta \to 0$ in position space is the same ultraviolet phenomenon as a divergence from integrating over arbitrarily large loop momentum in momentum space. The lecture insists on this early because every later estimate can be translated back and forth between the two pictures.

3.3 Scalar Fields, Dimensional Analysis, and the Massless Propagator

Having set up the general language, the lecture narrows to the first case that matters physically here: scalar fields. There are not many fundamental scalars in particle physics, but the Higgs boson is enough to make the problem urgent.

The argument begins in a very characteristic way: before computing anything in detail, let us ask what dimensional analysis already forces on us. For a scalar field ϕ, the simplest action is

$$S_\phi \sim \int d^4x\, (\partial_\mu \phi)^2. \tag{3.3}$$

The exact normalization is not important for the present argument. What matters is that in units with $\hbar = c = 1$, the action is dimensionless. Therefore

$$1 \sim [S_\phi] \sim [d^4x]\,[\partial_\mu \phi]^2 \tag{3.4}$$

$$\sim L^4 \left(L^{-1}[\phi]\right)^2. \tag{3.5}$$

Hence

$$[\phi] = L^{-1}. \tag{3.6}$$

The lecture then checks the other familiar terms one by one. If we add a mass term $m^2\phi^2$, dimensional consistency requires

$$[m] = L^{-1}. \tag{3.7}$$

If we add a quartic interaction $\lambda\phi^4$, then

$$[\lambda] = L^0, \tag{3.8}$$

so the quartic scalar coupling is dimensionless in four dimensions.

Now let us return to the propagator. In the massless case there is no scale in the problem other than the separation Δ. Translation invariance says that the propagator depends only on $x - y$, Lorentz invariance says that it depends only on the invariant interval, and dimensional analysis says that it must carry the dimensions of two scalar fields. There is therefore essentially no choice:

$$G_\phi(\Delta) \propto \frac{1}{\Delta^2}, \qquad \Delta^2 = \Delta_\mu \Delta^\mu. \tag{3.9}$$

Numerical factors are present, but no new scale is.

The lecture then asks what happens when the field has a mass. The point is not to write the exact massive propagator. The point is to understand its short-distance behavior. For this the dispersion relation is enough:

$$E = \sqrt{p^2 + m^2}, \qquad E = |p| \quad \text{for a massless particle.} \tag{3.10}$$

When $|p| \gg m$, the massive expression forgets the mass and approaches the massless one. Large momentum means short distance, so the short-distance propagator must likewise forget the mass.

At larger separations the mass suppresses propagation. The lecture writes this schematically as an exponential cutoff, so one may think of the massive scalar propagator as behaving roughly like

$$G_{\phi,m}(\Delta) \approx \frac{1}{\Delta^2} e^{-m|\Delta|}. \tag{3.11}$$

This should be read as a guide, not an exact formula. Its purpose is to make one point clear: at large separation the mass matters, but near $\Delta = 0$ the singularity is still the same $1/\Delta^2$.

3.4 The Simplest Scalar Self-Energy Diagram and the Fine-Tuning Problem

Now the lecture cashes out all of this preparatory work. We turn from the abstract question "what does a propagator look like?" to the concrete question "what does the simplest loop correction do to a scalar mass?"

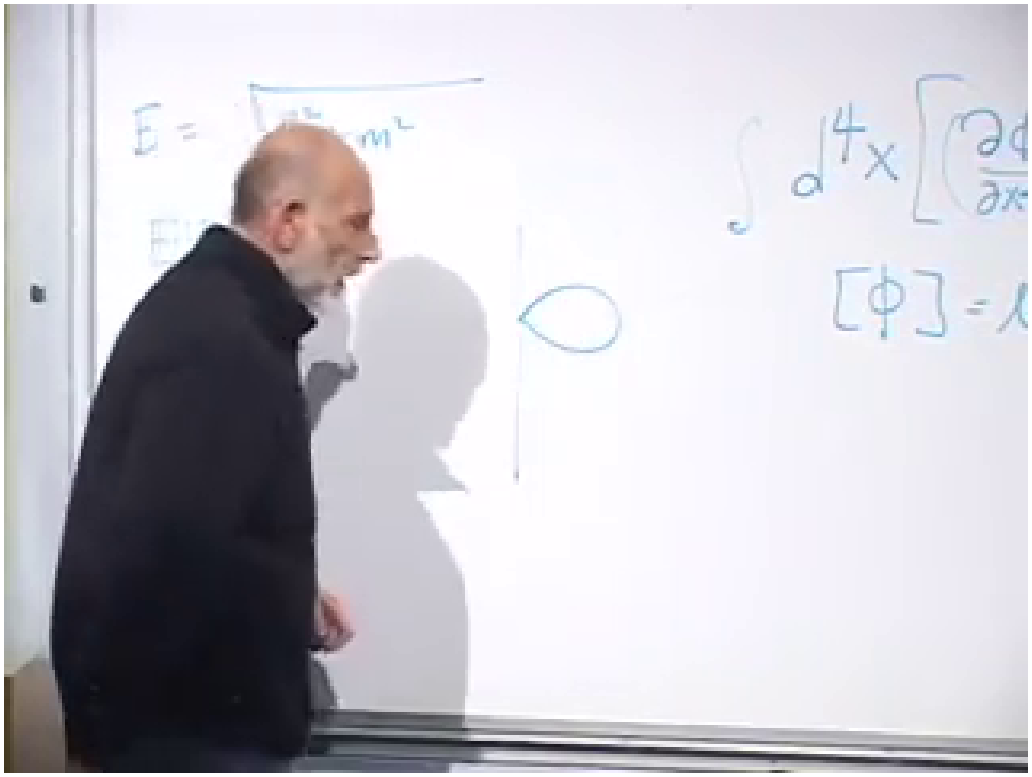

Figure 3.1: Blackboard moment at the transition from propagators and dimensional analysis to the simplest loop correction. The visible kinematic formulas and the partly visible action term are part of the same chain of reasoning as the loop diagram in the middle.

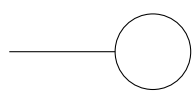

Figure 3.2: Minimal redraw of the central loop correction. The exact geometry on the board is slightly ambiguous, so the schematic form is kept deliberately spare.

The preserved blackboard frame is useful here because it shows the transition exactly as it happened in the lecture: the earlier discussion has not yet disappeared. One still sees the relativistic energy formula, its massless limit, part of the scalar-field action, and now in the middle the simplest loop correction.

Cleaned up, and with the partly visible action term written in the standard form suggested by the immediately preceding discussion, the board content may be summarized schematically as

$$E = \sqrt{p^2 + m^2}, \qquad E = |p|, \qquad S_\phi \sim \int d^4x\, (\partial_\mu \phi)^2, \qquad [\phi] = L^{-1}. \tag{3.12}$$

So the loop diagram is not appearing from nowhere. It is the direct consequence of what we have just learned about short-distance propagation.

In a $\lambda\phi^4$ theory this diagram is a correction to the scalar mass term. The lecture reads it precisely that way: it modifies the amplitude for a scalar particle to be absorbed at a point and re-emitted from that same point, which is exactly what the $m^2\phi^2$ term already does. Its size is therefore of

order

$$\delta m_\phi^2 \sim \lambda\, G_\phi(0). \tag{3.13}$$

But the short-distance form of the propagator is $G_\phi(\Delta) \sim 1/\Delta^2$. At coincident points that is singular. If we now admit what the lecture keeps insisting on—that our theory should not be trusted down to arbitrarily small distances—then we replace the ill-defined coincidence limit by a shortest-distance cutoff δ. The estimate becomes

$$\delta m_\phi^2 \sim \frac{\lambda}{\delta^2}. \tag{3.14}$$

This is the quadratic divergence.

3.4.1 Question & Answer

Question. Why does the simplest scalar loop correction behave like λ/δ^2, and why is that such a serious problem?

Answer. The diagram contributes one factor of the quartic coupling λ, and the rest of the diagram is just a scalar propagator evaluated at zero separation. Since the propagator behaves like $1/\Delta^2$ at short distance, the only cutoff-regulated estimate is $1/\delta^2$. The seriousness lies in the scale. If δ is extremely small, the correction is extremely large, typically much larger than the scalar mass one hoped to explain. A small low-energy scalar mass must then come from a delicate cancellation between a bare parameter and a huge ultraviolet correction. That is the fine-tuning problem.

The lecture then says what one should not miss: this is the Higgs problem, but it is first a scalar-field problem. The Higgs boson is simply the physically urgent scalar.

3.5 Fermions: Dirac Lagrangian, Field Dimension, and Short-Distance Propagator

Only now, after the scalar divergence has been made unmistakable, does the lecture pivot to fermions. The logic is parallel to the scalar case, and the parallelism is part of the point.

The Dirac action is written schematically as

$$S_\psi \sim \int d^4x\, \bar\psi\, \gamma^\mu \partial_\mu \psi, \tag{3.15}$$

with the mass term

$$\int d^4x\, m\, \bar\psi\psi. \tag{3.16}$$

The gamma matrices carry no dimensions, and there is only one derivative. So the kinetic term gives

$$1 \sim [S_\psi] \sim [d^4x]\,[\bar\psi]\,[\partial_\mu]\,[\psi] \tag{3.17}$$

$$\sim L^4 [\psi]^2 L^{-1}. \tag{3.18}$$

Therefore

$$[\psi] = L^{-3/2}. \tag{3.19}$$

This point is stated almost playfully in the lecture: the half-integer power looks odd, but fermions are full of halves. A student remarks that the mass term would furnish the same result more quickly. Susskind agrees, but keeps the kinetic term in front because it displays the structure more transparently. One may then check from the mass term that

$$[d^4x]\,[m]\,[\bar{\psi}\psi] \sim L^4 \cdot L^{-1} \cdot L^{-3} \sim 1, \tag{3.20}$$

so m really does have the dimension of mass.

Now repeat the propagator argument. The fermion propagator has the dimensions of two fermion fields, so in the massless case

$$S_\psi(\Delta) \sim \frac{1}{\Delta^3}. \tag{3.21}$$

This is the lecture's preferred shorthand for dimensional reasoning. If one restores the spinor structure that the lecture temporarily suppresses, then one inserts a gamma matrix and one power of the separation vector upstairs:

$$S_{ij}(\Delta) \propto \frac{(\gamma_\mu \Delta^\mu)_{ij}}{\Delta^4}. \tag{3.22}$$

That more structured form is useful as a consistency check, but for the ultraviolet counting the compact rule $1/\Delta^3$ is enough.

The lecture then makes exactly the same ultraviolet point as in the scalar case: a nonzero fermion mass suppresses propagation at large distance, but near $\Delta = 0$ the short-distance behavior is governed by the same massless estimate.

3.6 Closed Fermion Loops, Yukawa Couplings, and the Supersymmetry Motive

Now we come to the real turning point of the lecture. We have a bad scalar loop. Can a fermionic diagram cancel it?

Before estimating any such diagram, the lecture first settles the sign. This is not a secondary technicality. It is the first reason to think a cancellation could even exist.

3.6.1 Question & Answer

Question. Why does a closed fermion loop come with a minus sign, and how can that possibly help with the scalar divergence?

Answer. The lecture gives a neat exchange argument. Begin with two disconnected identical loops. Their product is positive no matter what sign each one-loop factor has. Now cut and reconnect them by exchanging the two identical particles. For bosons, exchange does nothing to the sign because the two-particle wavefunction is symmetric:

$$\Psi_B(x_1, x_2) = +\Psi_B(x_2, x_1). \tag{3.23}$$

For fermions, exchange produces a minus sign because the wavefunction is antisymmetric:

$$\Psi_F(x_1, x_2) = -\Psi_F(x_2, x_1). \tag{3.24}$$

But after the exchange, the reconnected object is just a one-loop diagram. Therefore the one-loop fermion diagram has the opposite sign from the corresponding bosonic loop. That still does not solve the problem. But it opens the only door through which the problem might be solved: a fermion loop can carry the same ultraviolet scaling with the opposite sign.

The simplest scalar–fermion coupling is a Yukawa term,

$$g\,\phi\,\bar{\psi}\psi. \tag{3.25}$$

Its dimension is fixed just as before:

$$1 \sim [d^4x]\,[g]\,[\phi]\,[\bar{\psi}\psi] \tag{3.26}$$

$$\sim L^4[g]L^{-1}L^{-3}, \tag{3.27}$$

so

$$[g] = L^0. \tag{3.28}$$

The Yukawa coupling is dimensionless.

Now estimate the loop correction. The scalar is absorbed at one point and re-emitted at another. There are two Yukawa vertices, hence a factor g^2, and two fermion propagators, each of order $1/\Delta^3$. Holding one point fixed and integrating over the separation of the other, the lecture arrives at

$$\mathcal{A}_f \sim -g^2 \int d^4\Delta\, \frac{1}{\Delta^6}. \tag{3.29}$$

The integral contributes four powers of length upstairs and six downstairs, so by dimensional analysis

$$\mathcal{A}_f \sim -\frac{g^2}{\delta^2}. \tag{3.30}$$

The integrand itself is positive; the overall minus sign is the closed-fermion-loop sign.

At this point the lecture makes an important sober observation. It does not yet look as though we have solved anything. We still have a number multiplying $1/\delta^2$. But now, at minimum, we have found a possible source of a minus sign. The scalar loop contributes

$$+\frac{\lambda}{\delta^2}, \tag{3.31}$$

while the fermion loop contributes

$$-\frac{g^2}{\delta^2}. \tag{3.32}$$

So a cancellation of the dangerous divergence becomes imaginable.

But it would be a miracle if nature simply chose the two couplings to satisfy

$$g^2 = \lambda \tag{3.33}$$

for no reason at all. And even that is not quite enough. For exact cancellation one would want the bosonic and fermionic propagators to be matched closely enough in the ultraviolet, which in practice means matching masses. If the masses differ, the full answers need not cancel exactly; but as the lecture emphasizes, the divergent part can still cancel provided the short-distance behavior remains sufficiently close.

Now we can finally say why supersymmetry enters. Supersymmetry is not yet introduced here as an abstract algebraic game. It is introduced as precisely the sort of mathematical structure one would want if one hoped to relate bosonic and fermionic couplings and masses strongly enough to remove the nasty scalar divergences. The formal mathematics is postponed to the next lecture. The motive has now been established.

3.7 Summary

The lecture begins by refusing to rush into supersymmetry. Instead it reopens the ultraviolet problem that makes supersymmetry worth wanting in the first place. We first recast loop divergences in both position-space and momentum-space language, then use dimensional analysis to determine the short-distance propagators for scalar and fermion fields. From there the central scalar self-energy estimate follows almost immediately: $\delta m_\phi^2 \sim \lambda/\delta^2$. That is the fine-tuning problem in its sharpest form.

Only after the problem is clear do fermions enter as possible rescuers. Their propagator scales as $1/\Delta^3$, a closed fermion loop comes with a minus sign, and the simplest Yukawa loop contributes $-g^2/\delta^2$. Thus the lecture does not yet give us supersymmetry, but it does give us the reason to seek it: a symmetry capable of tying bosons and fermions together so tightly that the dangerous ultraviolet sensitivity of scalar masses is no longer a numerical accident.

Chapter 4

Symmetry, Supersymmetry, and Grassmann Numbers

This lecture begins by stepping back before it goes forward. Before supersymmetry can be introduced as a new and very strange symmetry, we are first reminded what an ordinary symmetry is, how it is represented in quantum mechanics, how infinitesimal generators encode the structure of a continuous group, and why a genuine symmetry must leave the energy unchanged. Only after that long preparation does the lecture make its sharp turn: every symmetry discussed so far preserves bosons as bosons and fermions as fermions, whereas supersymmetry mixes the two. These companion notes follow Leonard Susskind's Lecture 4 closely and are curated by LazyingArt LLC.

4.1 Symmetries As Operations On States

A symmetry relates things. That is the lecture's starting point, and it is worth keeping it in exactly that elementary form. Two configurations may look different, but if one is obtained from the other by a symmetry transformation, then the symmetry ties them together and forces equalities between their physical properties. In particular, states related by symmetry generally have the same mass or the same energy.

This is why the lecture begins with an almost homely example: left-right symmetry. The point is not the example itself. The point is that symmetry is not a slogan about invariance; it is a relation among transformed configurations. Once that reminder is in place, the lecture moves immediately to the mathematics.

Quantum mechanically, a symmetry is represented by an operator acting on a state vector. We begin with a state $|\psi\rangle$, whatever physical system it may describe, and we apply a transformation U. The result is another state,

$$U|\psi\rangle, \tag{4.1}$$

which represents the transformed physical configuration. Rotation is the main example in the lecture, though translation is mentioned at once as an equally basic possibility.

The operator U must be unitary:

$$U^\dagger U = 1. \tag{4.2}$$

The surviving board image is partially blocked, so we should be cautious about pretending to read

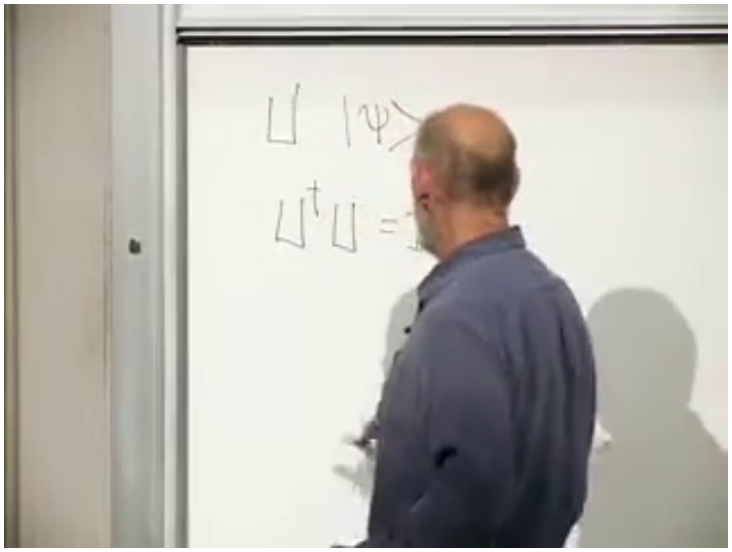

Figure 4.1: Early unitary-operator notation on the board. The upper line is partially occluded, but the transformed ket and the unitarity condition are both clear in intent.

more than is really visible. But the intended equation is secure, and the lecture explains at once what it means:

$$U^\dagger = U^{-1}. \tag{4.3}$$

This is the statement that norms are preserved, and hence that probabilities are preserved. In Susskind's phrasing, the vector must keep the same length in Hilbert space.

Before going further, the lecture makes a small but important remark. Supersymmetry does not fit very comfortably into the ordinary classification of symmetries as continuous or discrete. Precisely because of that, he first reviews the familiar continuous case carefully.

4.1.1 Question & Answer

Are we rotating ordinary space or Hilbert space? The lecture pauses here over a natural question. If a state vector lives in Hilbert space, then when we speak about rotation are we rotating ordinary three-dimensional space or are we rotating Hilbert space?

The answer is: both, but not in the same sense. The physical operation is a rotation in ordinary space. The mathematical representation of that physical operation is a unitary transformation acting in Hilbert space. So U is not "a rotation of states" in the naive geometric sense; it is a Hilbert-space operator that represents the act of rotating the physical system.

4.2 Infinitesimal Transformations And Generators

Once the operator U is on the table, the lecture narrows to very small transformations. This is the first genuinely technical step. No transformation at all is represented by

$$U = 1, \tag{4.4}$$

not by $U = 0$. So an infinitesimal transformation must differ from the identity by a small correction. If the small parameter is ϵ, then

$$U(\epsilon) = 1 + i\epsilon L + O(\epsilon^2). \tag{4.5}$$

To preserve unitarity at leading order, L must be Hermitian:

$$L^\dagger = L. \tag{4.6}$$

For rotations the generator must be labeled by a direction in space. Thus we have L_x, L_y, and L_z, corresponding to infinitesimal rotations about the three coordinate axes. A rotation about an arbitrary axis is then a linear combination of these three basic generators. In this sense there are only three independent infinitesimal rotational generators; every other small rotation is built from them.

At this stage the lecture is still speaking about ordinary symmetries, and in the case of rotations these generators are just angular momentum operators. But the notation L_i is deliberately more general, because the real point is not angular momentum itself. The point is that continuous symmetry is carried by infinitesimal generators.

4.3 Why Commutators Encode The Group

Now comes the long algebraic beat of the lecture. One of the most important facts about a continuous group is its commutator algebra. The lecture first states this abstractly and only then gives it geometric meaning.

For rotations one writes, first in components,

$$[L_x, L_y] = iL_z, \qquad [L_y, L_z] = iL_x, \qquad [L_z, L_x] = iL_y, \tag{4.7}$$

and then in condensed form,

$$[L_i, L_j] = i\epsilon_{ijk} L_k. \tag{4.8}$$

What matters is that the commutator of any two generators is again proportional to a generator. The set closes under commutation.

But the lecture does not want us to take that on authority. It wants us to see what a commutator means.

A small-rotation example. Take a small rotation about the x-axis and a small rotation about the y-axis:

$$U_x(\epsilon) = 1 + i\epsilon L_x, \qquad U_y(\delta) = 1 + i\delta L_y. \tag{4.9}$$

Now perform the loop

$$U_x(\epsilon)\, U_y(\delta)\, U_x^{-1}(\epsilon)\, U_y^{-1}(\delta). \tag{4.10}$$

A useful side remark, made in the lecture because a student interrupts to insist on it, is that unitarity lets us rewrite "dagger" as "inverse." Undoing a small transformation means replacing $+$ by $-$:

$$U_x^{-1}(\epsilon) = 1 - i\epsilon L_x, \qquad U_y^{-1}(\delta) = 1 - i\delta L_y. \tag{4.11}$$

If both rotations were about the same axis, the whole loop would collapse to the identity. The interesting case is that the axes are different. Then the first-order terms in ϵ and δ cancel, but the

mixed term survives. The spoken combinatorics become messy at this point, and the lecture briefly struggles with a factor of two before settling down. For the written notes it is best to keep the narrative struggle and typeset only the cleaned final result:

$$U_x(\epsilon)\, U_y(\delta)\, U_x^{-1}(\epsilon)\, U_y^{-1}(\delta) = 1 + i\epsilon\delta L_z + O(\epsilon^2, \delta^2). \tag{4.12}$$

So the failure of cancellation is itself another small rotation, here about the z-axis, by an angle $\epsilon\delta$. This is the geometric meaning of the commutator. The commutator keeps track of the non-cancellation of operations performed in different orders.

The lecture then broadens the discussion from rotations to translations. A small translation along the x-axis or y-axis is generated by momentum:

$$T_x(\epsilon) = 1 + i\epsilon P_x, \qquad T_y(\epsilon) = 1 + i\epsilon P_y. \tag{4.13}$$

Here the parameter ϵ is now a distance rather than an angle. The lecture also pauses to correct another possible misunderstanding: these are not literal motions of a system requiring forces and accelerations. They are mathematical transformations of a state, asking what the world would look like if everything were displaced.

Two translations commute. Translate in x, then in y, then undo x, then undo y, and one comes back to the same place. Therefore

$$[P_i, P_j] = 0. \tag{4.14}$$

Translations and rotations do not commute. Translate a point, rotate it, translate back along the original axis, and then undo the rotation, and one does not return to the original point. The lecture gives the corresponding algebra in rough spoken notation; the standard normalized form is

$$[L_i, P_j] = i\epsilon_{ijk} P_k. \tag{4.15}$$

The exact index bookkeeping is not the important thing here. What matters is the idea Susskind emphasizes repeatedly: the whole local structure of the group is encoded in these commutation relations. The commutator algebra is the multiplication table of the group written in infinitesimal form.

4.4 Symmetry And The Hamiltonian

Only after this algebraic recap does the lecture make its central dynamical pivot. A true symmetry should not change the energy of a state. That sounds obvious in words, but the lecture takes pains to derive its operator content carefully, and even runs through the derivation twice once a student asks for it again.

The first sparse board image in this sequence isolates the transformed energy eigenstate:

The object of interest is simply

$$U|E\rangle. \tag{4.16}$$

The second board image isolates the defining eigenvalue equation:

We start from

$$H|E\rangle = E|E\rangle. \tag{4.17}$$

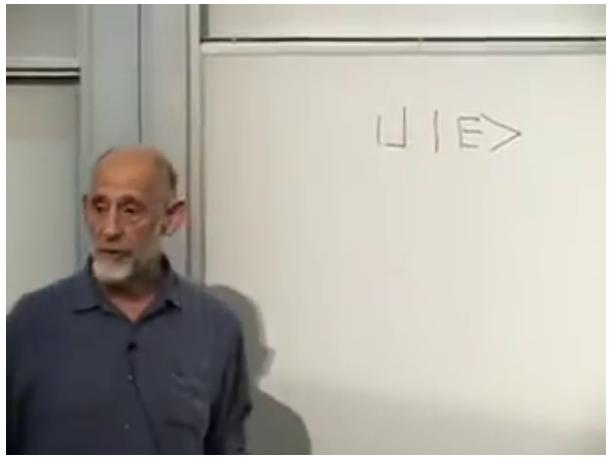

Figure 4.2: Symmetry operator acting on an energy eigenstate.

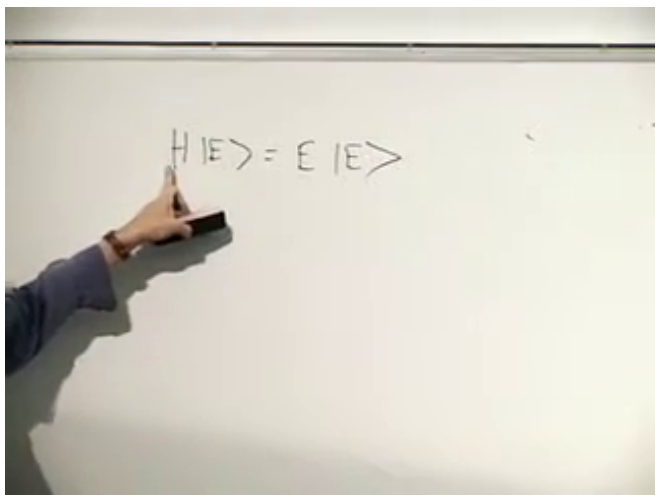

Figure 4.3: Hamiltonian eigenvalue equation.

The lecture uses E both as the scalar energy eigenvalue and as the label of the ket $|E\rangle$. We will keep the notation because it is the lecture's notation, but we must keep the roles distinct in words: the E multiplying the ket is a number, while $|E\rangle$ is the state.

Now suppose U is a symmetry. Then $U|E\rangle$ must have the same energy. That is the whole dynamical content of symmetry in this context. Therefore

$$H|E\rangle = E|E\rangle, \tag{4.18}$$
$$H\,U|E\rangle = E\,U|E\rangle, \tag{4.19}$$
$$= U\,E|E\rangle, \tag{4.20}$$
$$= U\,H|E\rangle. \tag{4.21}$$

The middle step is the one the lecture lingers over. Since E is just a scalar, it can be moved through the operator U. Then, because $E|E\rangle = H|E\rangle$, the scalar may be replaced by the Hamiltonian acting on the original eigenket.

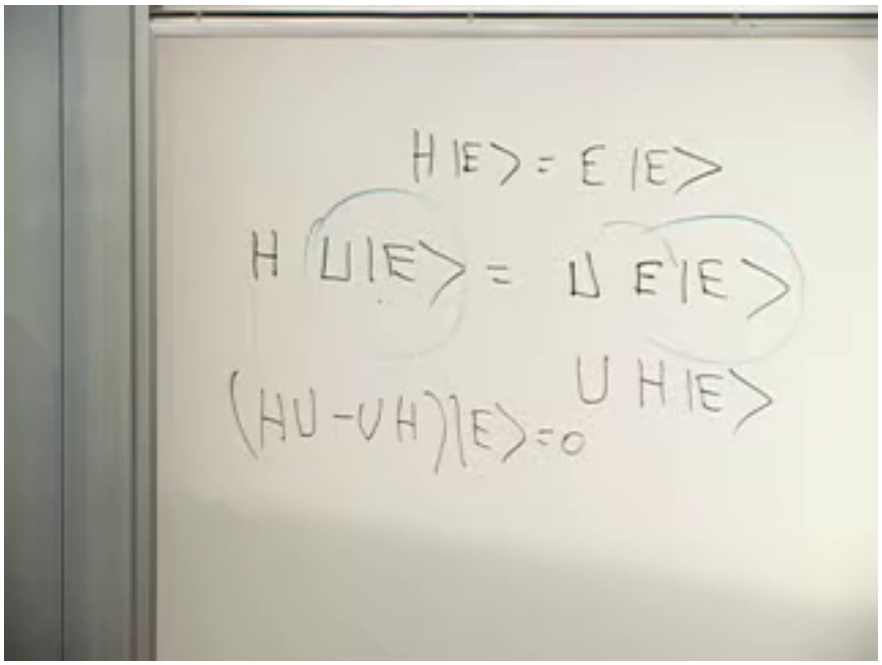

Figure 4.4: From the transformed eigenstate to a commutator condition. The curved board marks are part of the derivational staging, not decorative flourishes.

$$H|E\rangle = E|E\rangle$$

$$H\,U|E\rangle = E\,U|E\rangle = U\,E|E\rangle = U\,H|E\rangle$$

$$(HU - UH)|E\rangle = 0$$

Figure 4.5: A clean companion reconstruction of the board derivation; the curved strokes only echo the lecture's guide marks.

Subtracting the first and last expressions gives

$$(HU - UH)|E\rangle = 0. \tag{4.22}$$

Because that board layout is so important pedagogically, it is useful to keep a clean companion rendering nearby:

If this annihilates every energy eigenstate, and if the energy eigenstates form a complete basis, then the operator itself must vanish:

$$[U, H] = 0. \tag{4.23}$$

That is the mathematical criterion for a symmetry transformation.

Now return to the infinitesimal form

$$U(\epsilon) = 1 + i\epsilon L_i + O(\epsilon^2). \tag{4.24}$$

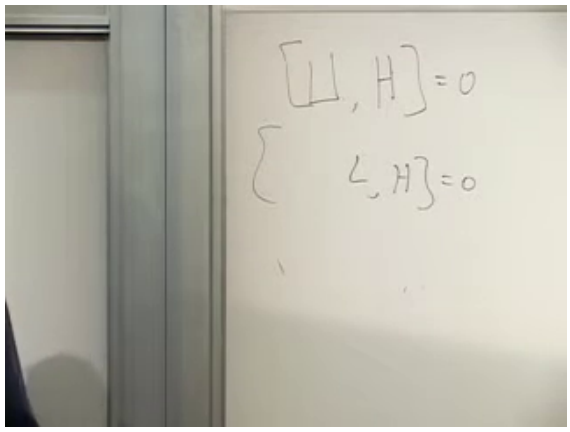

Figure 4.6: Symmetry operators and generators commuting with the Hamiltonian. The upper relation is fully legible; the lower one is standardized cautiously as $[L_i, H] = 0$.

Since the identity commutes with everything, the same statement tells us that the generators commute with the Hamiltonian:

$$[L_i, H] = 0. \tag{4.25}$$

The lower board label in the surviving frame is a little soft, but the normalized form $[L_i, H] = 0$ is the safest and most coherent way to write the lecture's conclusion.

This is more than a static condition. If an operator commutes with the Hamiltonian, then in the Heisenberg sense it is time-independent:

$$[O, H] = 0 \qquad \Longrightarrow \qquad \dot{O} = 0. \tag{4.26}$$

Thus symmetry and conservation law stand very close to each other.

The lecture immediately cashes this out in the familiar example of atomic states. Fix the total angular momentum l. The different magnetic quantum numbers m are rotated into one another, so they must all have the same energy:

$$E_{l,m} = E_l. \tag{4.27}$$

That is precisely the content of rotational symmetry, provided there is no external magnetic or electric field to spoil it.

4.4.1 Question & Answer

What counts as a true symmetry? This is where the lecture corrects a very common mistake. Suppose we rotate a spin in empty space. Its energy does not change. Good: rotation is a symmetry.

Now put that same spin in a magnetic field and rotate only the spin. Its energy changes. Does that mean rotation is not a symmetry? No. It means we did not perform the symmetry transformation. We rotated one part of the setup relative to another part that we held fixed. A genuine symmetry operation would rotate the spin and the magnetic field together.

The same point appears again for translation. An electron in the field of a proton does not keep the same energy if we translate only the electron. But that is not the symmetry operation either. Translating the whole electron-proton system is the symmetry. So when we say that symmetry

does not change the energy, we mean a transformation of the whole physical configuration, not an isolated manipulation of one subsystem inside a fixed background.

4.5 Why Supersymmetry Is Different

Only after the entire review of ordinary symmetry does the lecture introduce the new idea. All symmetries discussed so far have one common feature: they preserve particle type. Rotating an electron gives an electron. A color transformation turns a quark into another quark. A scalar goes to a scalar; a spin-$\frac{1}{2}$ particle goes to another spin-$\frac{1}{2}$ particle. The symmetries may mix internal labels, spatial orientations, or magnetic quantum numbers, but they do not turn bosons into fermions.

Supersymmetry breaks exactly that pattern. It is a symmetry whose generators take bosons into fermions and fermions into bosons. The lecture presents this as a real escalation, not as one more example on the same list. Ordinary symmetry multiplets are already familiar: rotational symmetry forces the states with different m-values inside a fixed l-multiplet to have the same energy. Supersymmetry proposes a new kind of multiplet, one that crosses the boson-fermion divide.

The physical motivation is also kept in view. If a theory truly has supersymmetry, then bosons and fermions come in matched partners of equal mass. That is precisely the structure one would want if bosonic and fermionic contributions are to cancel the large renormalization effects discussed earlier in the course. So the lecture does not motivate supersymmetry by elegance alone. It motivates it by the possibility of a dynamical payoff.

4.6 Supersymmetry Generators And Statistics

The lecture now asks the central constructive question: what sort of operator could take a boson into a fermion and a fermion into a boson?

Let A and $A^\dagger$ be bosonic annihilation and creation operators, and let C and $C^\dagger$ be the corresponding fermionic operators. Then a simple prototype is

$$Q = C^\dagger A + A^\dagger C. \tag{4.28}$$

The first term annihilates a boson and creates a fermion. The second annihilates a fermion and creates a boson. Schematically,

$$Q|B\rangle \sim |F\rangle, \qquad Q|F\rangle \sim |B\rangle. \tag{4.29}$$

A student asks whether this means we are somehow creating a mysterious boson-fermion superposition as a single state. The lecture corrects that immediately. Q is an operator, not a state. Acting on a general superposition it may of course produce another superposition, but that is just the ordinary linearity of operators. The distinctive property is simpler: acting on a boson it gives a fermion, and acting on a fermion it gives a boson.

4.6.1 Question & Answer

What happens when Q acts on the vacuum? This is the natural place to make a mistake, so the lecture stops and checks it explicitly.

Apply

$$Q = C^\dagger A + A^\dagger C \tag{4.30}$$

to the vacuum $|0\rangle$. The operator A tries to annihilate a boson, but no boson is present. The operator C tries to annihilate a fermion, but no fermion is present either. Therefore

$$Q|0\rangle = 0. \tag{4.31}$$

This is not the vacuum again. It is the zero vector. The lecture is explicit about that distinction.

The next question is about spin. If a spin-zero boson is mapped to a fermion, which fermion appears? A spin-$\frac{1}{2}$ fermion has two components. So a single unindexed generator cannot be enough. The lecture's answer is that supersymmetry generators must carry a spinor index:

$$Q_\alpha. \tag{4.32}$$

Schematically one should think of

$$Q_\alpha|B\rangle \sim |F_\alpha\rangle, \qquad Q_\alpha|F_\beta\rangle \sim \delta_{\alpha\beta}|B\rangle. \tag{4.33}$$

The precise coefficients are not the point here. The point is that the generator itself carries half-integer spin because it changes spin by half a unit.

This already marks supersymmetry out as different from ordinary observables. Supersymmetry generators are fermionic operators: they contain an odd number of fermion operators. That is exactly why they can change bosons into fermions and fermions into bosons.

To see how radical this is, the lecture reviews the algebra of bosonic and fermionic creation and annihilation operators. For bosons we normalize the relations as

$$[A_i, A_j] = 0, \qquad [A_i^\dagger, A_j^\dagger] = 0, \qquad [A_i, A_j^\dagger] = \delta_{ij}. \tag{4.34}$$

Hence bosonic creation operators commute:

$$A_i^\dagger A_j^\dagger = A_j^\dagger A_i^\dagger. \tag{4.35}$$

So a two-boson state is symmetric under interchange of the labels i and j.

For fermions we replace commutators by anticommutators:

$$\{C_i, C_j\} = 0, \qquad \{C_i^\dagger, C_j^\dagger\} = 0, \qquad \{C_i, C_j^\dagger\} = \delta_{ij}. \tag{4.36}$$

In particular,

$$(C^\dagger)^2 = 0, \tag{4.37}$$

which is the Pauli exclusion principle in operator form: two fermions cannot occupy the same state. And for distinct labels,

$$C_i^\dagger C_j^\dagger = -\,C_j^\dagger C_i^\dagger. \tag{4.38}$$

So fermionic two-particle states are antisymmetric under exchange.

This is the point at which the lecture broadens again. Bosonic field amplitudes can become large and classical-looking. Fermionic objects cannot, because their squares vanish. That is what forces us toward a new kind of arithmetic.

4.7 Grassmann Numbers

The lecture introduces Grassmann numbers as the bookkeeping language appropriate to fermionic quantities and, in particular, to supersymmetry generators. Ordinary numbers commute:

$$\alpha_i \alpha_j = \alpha_j \alpha_i. \tag{4.39}$$

Grassmann numbers anticommute:

$$\theta_i \theta_j = -\theta_j \theta_i. \tag{4.40}$$

Setting $i = j$ gives

$$\theta_i^2 = 0. \tag{4.41}$$

And ordinary numbers commute with Grassmann numbers:

$$\alpha\theta = \theta\alpha. \tag{4.42}$$

The lecture is careful not to overinterpret these objects. They are not directly measurable quantities. They are a new algebraic device. It is also careful about parity. A single θ is odd. A product such as $\theta_1\theta_2$ is even, in the graded sense that it commutes with other even quantities, but it need not be an ordinary scalar in the naive sense, because it may still be nilpotent.

With one Grassmann variable θ, the most general function is

$$f(\theta) = \alpha + \beta\theta. \tag{4.43}$$

There is no quadratic term because $\theta^2 = 0$. With two Grassmann variables,

$$f(\theta_1, \theta_2) = \alpha + \beta_1\theta_1 + \beta_2\theta_2 + \gamma\theta_1\theta_2. \tag{4.44}$$

That is already the most general polynomial. The lecture lingers over how little room there is in this algebra.

Differentiation is then introduced with just one twist. For one variable we define

$$\partial_\theta(\alpha + \beta\theta) = \beta. \tag{4.45}$$

For two variables we obtain

$$\partial_{\theta_1} f = \beta_1 + \gamma\theta_2, \qquad \partial_{\theta_2} f = \beta_2 - \gamma\theta_1. \tag{4.46}$$

The minus sign in the second formula is the whole point. The derivative operator with respect to a Grassmann variable is itself Grassmann-odd, so when it passes through another Grassmann quantity it picks up a sign. The lecture emphasizes that this is precisely what is needed to preserve the usual product rule of calculus.

The one-variable exponential remains simple:

$$e^{a\theta} = 1 + a\theta. \tag{4.47}$$

Therefore

$$e^{a\theta} e^{b\theta} = e^{(a+b)\theta}. \tag{4.48}$$

Up to this point the algebra behaves exactly as the lecture hopes.

But then the lecture tries two Grassmann variables and the tone changes from exposition to investigation. One certainly computes

$$e^{\theta_1} e^{\theta_2} = (1 + \theta_1)(1 + \theta_2) = 1 + \theta_1 + \theta_2 + \theta_1\theta_2. \tag{4.49}$$

The tempting next step is to compare this with $e^{\theta_1+\theta_2}$. Here the lecture runs into a real puzzle. Because $(\theta_1 + \theta_2)^2$ brings in $\theta_1\theta_2 + \theta_2\theta_1$, the naive exponential law becomes slippery. The discussion does not settle the matter. It is important not to flatten this into a polished theorem that the lecture itself never reached. The honest endpoint is that Grassmann arithmetic exists, is useful, and is essential, but the multi-variable exponential is left here as an unresolved question.

The lecture closes this part by noting that integration over Grassmann variables can also be defined, and that because the space of functions is so small the whole integration table will be correspondingly short. But that development is postponed. What matters for now is the existence of a consistent algebraic framework in which fermion operators and supersymmetry generators naturally live.

4.8 Summary

The lecture unfolds in a very deliberate order. We first rebuild the ordinary notion of symmetry: a unitary operator acting on states, with infinitesimal generators that close under commutation. We then learn to read the whole local structure of a continuous group from those commutators. Next comes the dynamical condition that turns transformation theory into symmetry theory:

$$[U, H] = 0, \qquad [L_i, H] = 0. \tag{4.50}$$

Only after that groundwork is laid does supersymmetry appear, not as one more ordinary symmetry, but as a new sort of transformation that mixes bosons and fermions. That step forces us beyond ordinary commuting quantities and into Grassmann arithmetic. The lecture therefore ends, quite appropriately, not with a closed formal system, but with the sense that a new algebra has opened up and that supersymmetry lives at its center.